



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-6

Vacuum Cleaner Problem

Aim

To write a Python program to simulate the Vacuum Cleaner Problem.

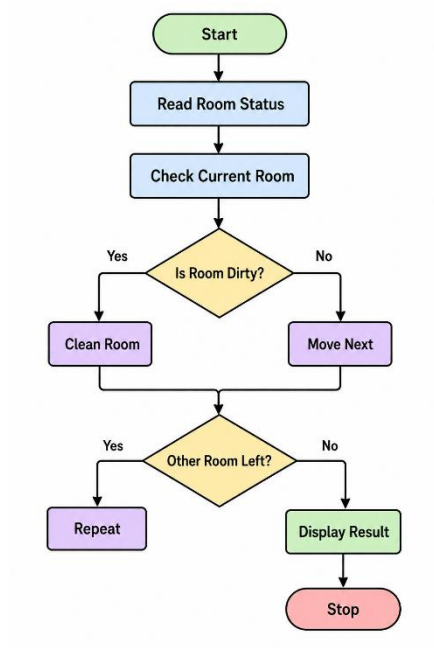
Objective

To clean all dirty rooms using a simple vacuum cleaner agent.

Algorithm

1. Start.
2. Read the status of Room A and Room B.
3. Place the vacuum in the starting room.
4. If the current room is dirty, clean it.
5. Move to the other room and repeat.
6. Display the final status.
7. Stop.

Flowchart



Code

```
roomA = input("Enter status of Room A (Dirty/Clean): ").lower()
roomB = input("Enter status of Room B (Dirty/Clean): ").lower()

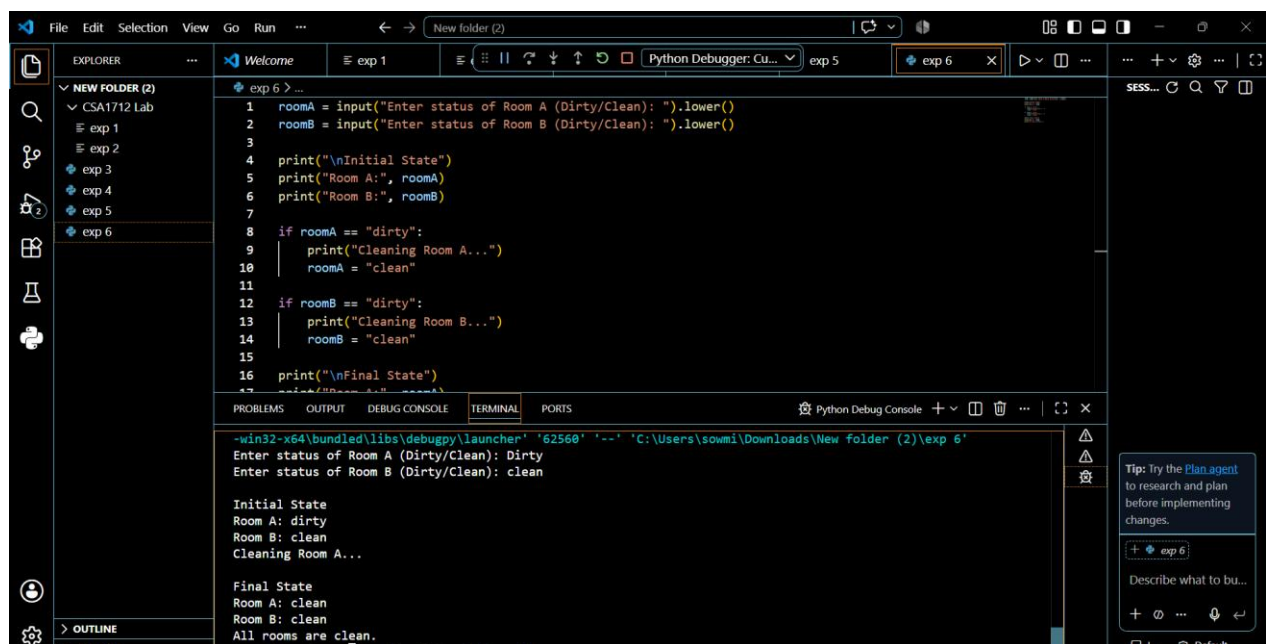
print("\nInitial State")
print("Room A:", roomA)
print("Room B:", roomB)

if roomA == "dirty":
    print("Cleaning Room A...")
    roomA = "clean"
```

```
if roomB == "dirty":  
    print("Cleaning Room B...")  
    roomB = "clean"
```

```
print("\nFinal State")  
print("Room A:", roomA)  
print("Room B:", roomB)  
print("All rooms are clean.")
```

Output



The screenshot shows a Python IDE with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The code in the editor is a Python script for a vacuum cleaner simulation. The terminal shows the execution of the script, including user input for room status and the resulting state of the rooms.

```
1 roomA = input("Enter status of Room A (Dirty/Clean): ").lower()  
2 roomB = input("Enter status of Room B (Dirty/Clean): ").lower()  
3  
4 print("\nInitial State")  
5 print("Room A:", roomA)  
6 print("Room B:", roomB)  
7  
8 if roomA == "dirty":  
9     print("Cleaning Room A...")  
10    roomA = "clean"  
11  
12 if roomB == "dirty":  
13     print("Cleaning Room B...")  
14     roomB = "clean"  
15  
16 print("\nFinal State")  
17
```

Terminal Output:

```
-win32-x64\bundled\libs\debugpy\launcher '62560' '-' 'C:\Users\sowmi\Downloads\New folder (2)\exp 6'  
Enter status of Room A (Dirty/Clean): Dirty  
Enter status of Room B (Dirty/Clean): clean  
  
Initial State  
Room A: dirty  
Room B: clean  
Cleaning Room A...  
  
Final State  
Room A: clean  
Room B: clean  
All rooms are clean.
```

Result

The vacuum cleaner agent cleaned all dirty rooms successfully.

Conclusion

The program successfully simulates the Vacuum Cleaner Problem using a simple intelligent agent.